Solution:

The first thing is to number the population(8585 children). The numbering has already been provided by the frequency table. There are 2 children with height of 105 cm, therefore we assign number 1 and 2 to the children those in the group 105 cm, number 3 to 6 is assigned to those in the group 107 cm and similarly all other children are assigned the numbers. In the last group 145 cms there are two children with assigned number 8584 and 8585.

Height (cm.)	Number of children	Cumulative Frequency
105	2	2
107	4	6
109	14	20
111	41	61
113	83	144
115	169	313
117	394	707
119	669	1376
121	990	2366
123	1223	3589
125	1329	4918
127	1230	6148
129	1063	7211
131	646	7857
133	392	8249
135	202	8451
137	79	8530
139	32	8562
141	16	8578
143	5	8583
145	2	8585
Total	8585	

Now we take 10 samples from the tables, since the population size is in 4 digits we can use the given random number table. Select the10 random numbers from 1 to 8585 in the table, Here, we consider column wise selection of random numbers, starting from first column.

Tippet's random number table							
2952	6641	3992	9792	7969	5911	3170	5624
4167	9524	1545	1396	7203	5356	1300	2693
2670	7483	3408	2762	3563	1089	6913	7991
0560	5246	1112	6107	6008	8125	4233	8776
2754	9143	1405	9025	7002	6111	8816	6446

The children with assigned number 2952 is selected and then see the cumulative frequency table where 2952 is present, now select the corresponding row height which is 123 cm, similarly all the selected random numbers are considered for the selection of the child with their corresponding height. The following table shows all the selected 10 children with their heights.

Child with assigned Number	2952	4167	2670	0560	2754
Corresponding Height (cms)	123	125	123	117	123
Child with assigned Number	6641	7483	5246	3992	1545
Corresponding Height (cms)	129	131	127	125	121

Example 8.4

Using the following random number table (Kendall-Babington Smith)

23	15	75	48	59	01	83	72	59	93	76	24	97	08	86	95	23	03	67	44
05	54	55	50	43	10	53	74	35	08	90	61	18	37	44	10	96	22	13	43
14	87	16	03	50	32	40	43	62	23	50	05	10	03	22	11	54	36	08	34
38	97	67	49	51	94	05	17	58	53	78	80	59	01	94	32	42	87	16	95
97	31	26	17	18	99	75	53	08	70	94	25	12	58	41	54	88	21	05	13

Draw a random sample of 10 four- figure numbers starting from 1550 to 8000.

XII Std - Business Maths & Stat EM Chapter 8.indd 180 2/29/2020 11:40:03 AM